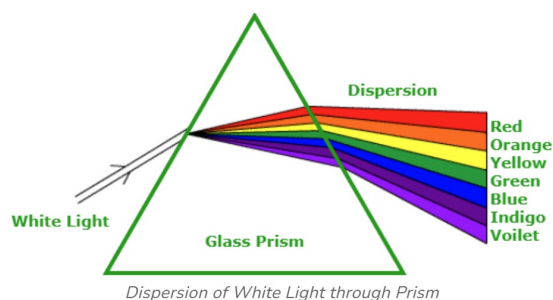


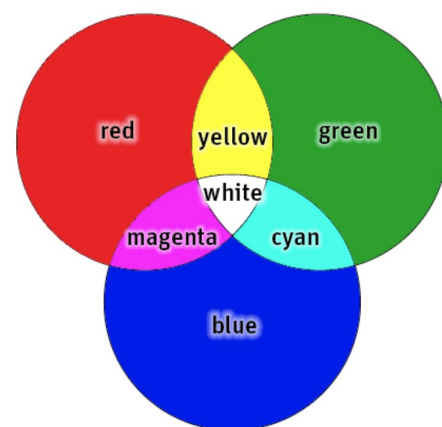
Light and Colours

- Visible Spectrum
 - White light is a **mixture** of different colours.

- When white light passes through a prism, it spreads into a **visible spectrum** or **colour spectrum**.



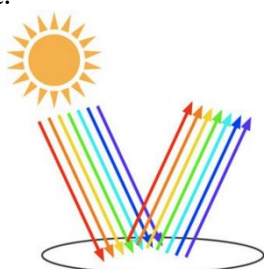
- Primary Colours and Secondary Colours
 - The primary colours of visible light are **red, green and blue**. All colours can be produced by mixing these three primary colours of light in different proportions.
 - Any two of the three primary colors of light produce the secondary colors of light. The secondary colors of light are cyan yellow (red + green), cyan (green + blue) and magenta (blue + red).



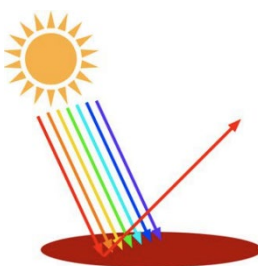
- Colours of objects in coloured light

- The colour of an object is determined by the colours of light reflected by the object.

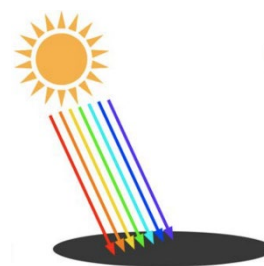
- A white object reflects all colours of light. The reflected colours of light are the same as the incoming light.



- A coloured object only reflects its own colours and absorbs all other colours.



- A dark object absorbs all colours of light.

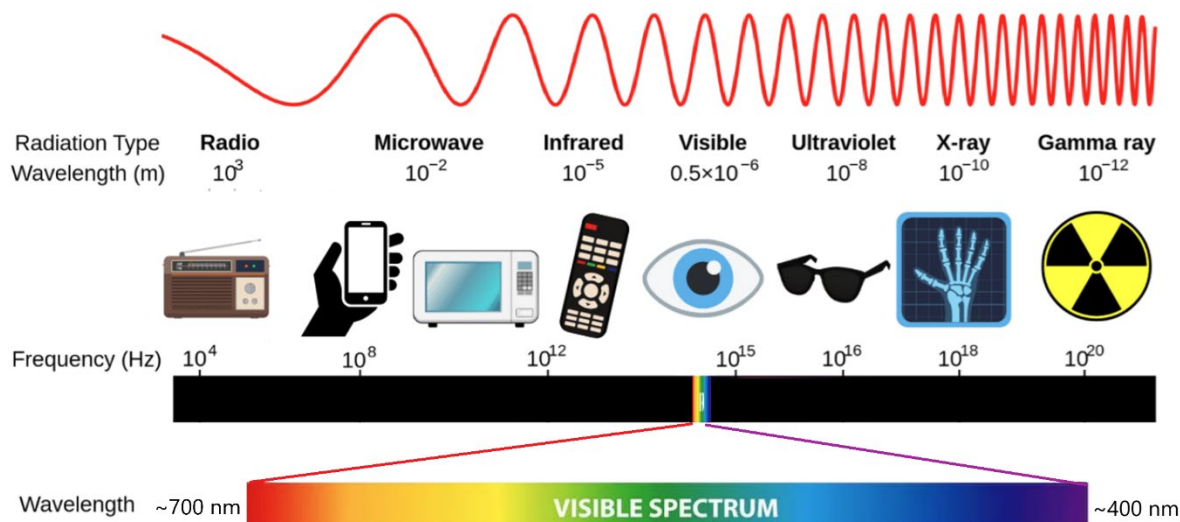


Complete the following table with colour of light (white light, primary colours of light or secondary colours of light).

Incident colour (s) of light	The object is appeared to be	Reflected colour of light
White	White	White
Red		red
Cyan		cyan
White	magenta	magenta
Red, yellow		red
Green		black
Magenta	Red + Green = yellow	red
cyan		green
Blue		dark

Electromagnetic Spectrum

- Electromagnetic spectrum consists of visible spectrum and invisible spectrum.

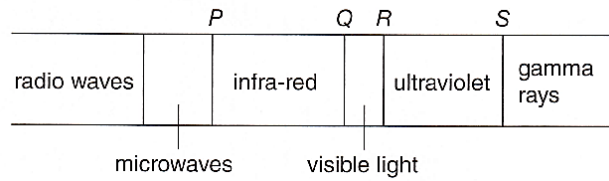


- They are **transverse waves**.
- They **carry energy**.
- They do not require a medium to travel and **can travel in a vacuum**.
- In a *vacuum*, their **speed c** is fixed at $3 \times 10^8 \text{ m s}^{-1}$. In a material, they travel slower at *various* speeds.
- They obey **$v = f\lambda$** .
- They show **reflection, refraction, diffraction and interference**.

	Source	Detector	Uses	Wavelength
Radio waves	Radio and TV transmitter	Antenna	- Radio/TV broadcasting - Mobile phone communication	$10^{-1} - 10^4 \text{ m}$
Microwaves	Microwave transmitter	Microwave receiver	- Microwave oven - Satellite communication - Radar	$10^{-3} - 10^{-1} \text{ m}$
Infrared	Objects above 0 K	Skin, phototransistor	- Remote control used in TV - Auto-focus of camera - Infra-red thermometer - Radiant heater,	$10^{-6} - 10^{-3} \text{ m}$
Visible light	Hot objects, Sun	Human eye	Lighting	$4 \times 10^{-7} - 7 \times 10^{-7} \text{ m}$
Ultraviolet	Very Hot objects, UV lamp, Sun	Fluorescent material	- To sterilize drinking water - Checking banknotes - Produce Vitamin D (Skin)	$10^{-9} - 10^{-8} \text{ m}$
X-ray	X-ray tube	Photographic film	- Production of see-through images (human body or luggage) - CT scan	10^{-10} m
Gamma ray	Radioactive substance	GM tube, photographic film	- Medical uses - Sterilization (killing bacteria in food) - Radiotherapy	10^{-12} m

1. Student draws the electromagnetic spectrum. However, the part for X-ray is missing. Where should the part for X-ray be?

- A. *P*
- B. *Q*
- C. *R*
- D. *S*



D

2. Which of the following shows the correct order of five electromagnetic waves arranged in the order of decreasing frequency?

- A X-rays, infra-red, microwaves, radio waves, ultraviolet
- B Ultra-violet, infra-red, radio waves, X-rays, microwaves
- C Radio waves, microwaves, ultraviolet, infra-red, X-rays
- D X-rays, ultraviolet, infra-red, microwaves, radio waves

D

3. Which of the following statements about electromagnetic waves is/are correct?

- (1) The speeds of electromagnetic waves are equal to the speed of light.
- (2) All kinds of electromagnetic waves can travel in a vacuum.
- (3) Electromagnetic waves are transverse waves.

- A (1) only
- B (2) only
- C (1) and (3) only
- D (1), (2) and (3)

D

4. Which of the following is not an application of microwaves?

- A. Radar
- B. Microwave oven
- C. Satellite communication
- D. Thermometer

D

5. Which of the following is not an application of infrared radiation?

- A. Thermometer
- B. Auto-focus camera
- C. Laser pointer
- D. Electric oven

C

6. Which of the following type of electromagnetic waves is best used for killing cancer cells?

- A. Gamma rays
- B. Infrared radiation
- C. Visible light
- D. Ultraviolet radiation

A

7. Which of the following correctly shows the uses of microwaves, radio waves and ultraviolet radiation?

<u>Microwaves</u>	<u>Radio waves</u>	<u>Ultraviolet radiation</u>
A. Radar	Burglar alarm	Medical diagnosis
B. Communication	TV broadcast	Checking banknote
C. Tracer	Sterilizing water	Remote control
D. Cooking	Radiotherapy	Thermography

B

8. Given the following types of electromagnetic waves

- (1) radio waves
- (2) yellow light
- (3) green light

The waves listed in ascending order of their wavelength are

- A. (1), (3), (2)
- B. (2), (1), (3)
- C. (3), (1), (2)
- D. (3), (2), (1)

D

9. Arrange the following electromagnetic waves in descending order of their wavelength.

- (1) visible light
- (2) X-rays
- (3) radio waves

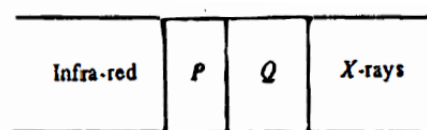
- A. (1), (3), (2)
- B. (2), (3), (1)
- C. (3), (1), (2)
- D. (3), (2), (1)

C

10. The diagram shows part of the electromagnetic spectrum. Which of the following statements is/are true?

- (1) The wavelength of P is longer than that of Q .
- (2) The velocity of P in a vacuum is smaller than that of Q .
- (3) Q can be deflected by an electric field.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only



A

11. (1) *X*-rays
(2) Ultra-violet rays.
(3) Radio waves

Arrange the above electromagnetic waves in ascending order of frequencies.

- A. (1) 、 (3) 、 (2)
B. (2) 、 (1) 、 (3)
C. (3) 、 (1) 、 (2)
D. (3) 、 (2) 、 (1)

D

12. Which of the following statements concerning infra-red radiation is correct?

- A. Infra-red can be detected by a Geiger-Muller counter.
B. Infra-red can be used to sterilizing drinking water.
C. Infra-red is a longitudinal wave.
D. Warm objects emit infra-red.

D

13. Part of the electromagnetic spectrum is shown above. Which of the following statements is/are correct?

- (1) *P* is ultra-violet and *Q* is infra-red.
(2) The wavelength of *P* is shorter than that of *Q*.
(3) The speed of *P* in vacuum is higher than that of *Q*.

- A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

<i>X</i> -rays	<i>P</i>	Visible light	<i>Q</i>	Microwaves
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C

14. Which of the following is not an application of the corresponding electromagnetic wave?

- | Electromagnetic wave | Application |
|----------------------|--|
| A. Ultra-violet | Camera autofocusing |
| B. Infra-red | Detecting survivors buried in landslides |
| C. Microwaves | Satellite communication |
| D. <i>X</i> -rays | Detecting weapons hidden in suitcases |

A

15. Which of the following is an application of ultra-violet radiation?

- A. Detecting cracks in railway tracks
B. Detecting survivors buried in landslides
C. Measurement of the depth of the sea-bed
D. Sterilization of drinking water

D

16. Which of the following is/are the reason(s) for not over-using ultra-violet lamps to produce a suntan effect?

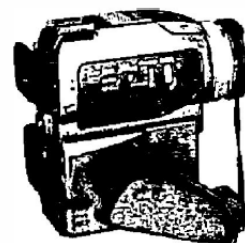
B

- (1) Ultra-violet radiation is highly penetrating and will damage body tissue.
 - (2) Ultra-violet radiation has a strong heating effect and will damage body tissue.
 - (3) Over exposure to ultra-violet radiation may cause skin cancer.
- A. (2) only
 - B. (3) only
 - C. (1) and (2) only
 - D. (1) and (3) only

17. The photograph shows a digital video camera recorder. The recorder has a nightshot function for capturing images in the dark. Which of the following electromagnetic waves is employed by the recorder in capturing images in the dark?

B

- A. radio waves
- B. infra-red
- C. ultra-violet
- D. X-rays



18. Which of the following shows the correct order of the relative positions of five electromagnetic waves in the electromagnetic spectrum, in the order of decreasing wavelength?

A

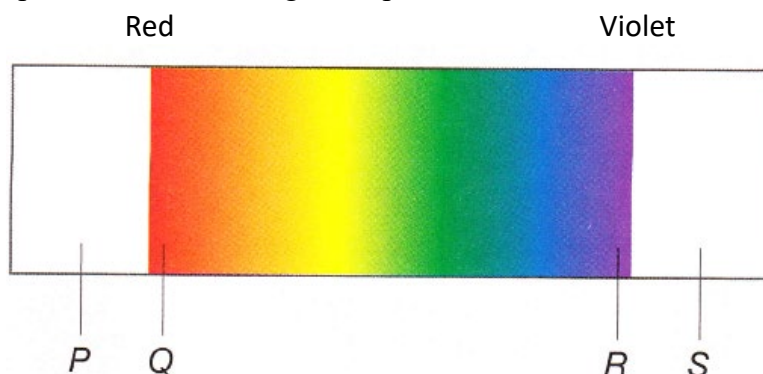
- A. microwaves, infra-red, visible light, ultra-violet, X-rays
- B. microwaves, ultra-violet, visible light, infra-red, X-rays
- C. X-rays, infra-red, visible light, ultra-violet, microwaves
- D. X-rays, ultra-violet, visible light, infra-red, microwaves

19. Which of the following phenomena demonstrate(s) that light is an electromagnetic wave?

B

- (1) Light bends when it travels across a boundary from one medium into another.
 - (2) Light reflects when it meets a polished metal surface.
 - (3) Light can travel from the Sun to the Earth.
- A. (1) only
 - B. (3) only
 - C. (1) and (2) only
 - D. (2) and (3) only

20. Figure below shows part of the electromagnetic spectrum.



- Which one has the longest wavelength, P, Q, R or S? **P**
- Which one has the highest speed in a vacuum? **Same**
- What is the speed of Q in a vacuum? **$3 \times 10^8 \text{ m s}^{-1}$**
- Name P and S.

P: infra-red radiation

S: ultra-violet radiation

21. The electromagnetic spectrum is shown in the figure.

radio waves	<i>X</i>	infra-red radiations	visible light	<i>Y</i>	X-rays	<i>Z</i>
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- Name the parts *X*, *Y* and *Z*.
- Which part of the spectrum has the shortest wavelength?
 - Which part of the spectrum has the lowest frequency?
- State one application of *Z*.

- X*: microwaves↵

Y: ultra-violet radiation↵

Z: gamma rays↵
- Gamma rays↵
 - Radio waves↵
- Any one of the following:↵
Radiotherapy↵
Sterilization↵
Medical imaging↵
(Or other reasonable answer)↵